An Information Bottleneck in Pairwise Neural Distance Estimation: From Simulation Success to Biological Data

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Abstract

Phylogenetic reconstruction infers evolutionary relationships among taxa from molecular sequences; distance-based methods build a tree from estimated pairwise distances. Neural estimators can be highly accurate on simulated data, but their empirical generalization remains incompletely characterized. On 1.71 million in-distribution simulated pairs, our column-wise Transformer reduces mean squared error by a factor of 2.6 relative to IQ-TREE. Yet across 418 mammalian and arthropod sub-alignments, Jukes–Cantor (JC) agrees more closely than Phyloformer, FIREFLY, and seven pair-encoder variants with per-MSA IQ-TREE GTR+Γ_4 distance estimates and yields more concordant tree topologies. These estimates are model-based references, not observable truth. An exact decomposition of the squared-loss risk splits this gap into two terms: an information ceiling (how much the pairwise input can reveal) and a shift between the simulated and empirical distributions. Because the empirical references are themselves model-based, we add a controlled test on a codon dataset with exact, known distances: on the non-reversible processes we test, encoders trained on mismatched GTR simulations still trail JC, while matched-simulation training reverses this comparison. A best-case predictor from pairwise features alone explains only part of the variance in true distances, and adding alignment-level context raises that share substantially. Two classical results explain why this is difficult: without an external rate calibration, the rate of evolution cannot be separated from elapsed time, while JC's mismatch fraction already ranks true distances correctly under any fixed time-reversible model. Together, these results point to two contributing causes: limited pairwise information and simulation-to-reality mismatch, without identifying either as the sole cause or ruling out model-capacity effects. Richer alignment-level context and better-matched simulation are two distinct routes forward; whether either closes the gap on real data remains open.

1 Introduction

Distance-based phylogenetics constructs a tree from a matrix of pairwise evolutionary distances. Given accurate distances, the Neighbor-Joining (NJ) algorithm (Saitou and Nei 1987) recovers the correct topology in polynomial time, with formal guarantees when estimation error is bounded (Atteson 1999). The quality of the inferred tree therefore reduces to the quality of the estimated distances, and for fifty years, the standard approach has been to assume a parametric substitution model (e.g. JC (Jukes and Cantor 1969), HKY (Hasegawa, Kishino, and Yano 1985), or GTR (Tavaré 1986)) and invert the observed sequence divergence through the model's transition probabilities. These models make likelihood computation tractable by assuming stationarity, reversibility, and homogeneity (SRH): a fixed equilibrium composition, detailed balance, and a common substitution process across lineages. In real data these assumptions may be violated—by lineage-specific biases, GC-biased gene conversion, selection shifts, and CpG hypermutation (Bird 1980; Foster 2004)—which can bias distance estimates and inferred topology (Galtier and Gouy 1995). Simulation-trained estimators can instead learn distances under richer processes, provided the simulations capture that heterogeneity. Deep learning (DL) is attractive because forward simulation under complex evolutionary models is inexpensive, yielding millions of labeled training pairs without requiring a closed-form distance correction. A growing body of work has pursued this idea. Liu et al. (2020) introduced convolutional distance estimators. Nesterenko et al. (2025) proposed Phyloformer, which processes entire multiple sequence alignments (MSAs) through axial self-attention. Rosenzweig and Hahn (2025) showed that compact S-networks suffice to approximate classical distance functions on simulated data. Broader neural-task context appears in Appendix H.1.

On simulated benchmarks, DL methods can outperform classical baselines on selected simulated metrics. Yet most report limited or no real-data evaluation (Braichenko, Borges, and Kosiol 2025; Zhu, Wang, and Yang 2025), leaving the magnitude and causes of the sim-to-real gap open.

This paper quantifies this gap with a column-wise pair-encoder Transformer (664k parameters) and seven training variants spanning three losses and two sampling schemes. We compare this baseline method with Phyloformer[1] and FIREFLY (Gao, Lee, and Liu 2026), as well as classical approaches under JC and GTR+Γ_4 models. Evaluation spans five regimes: a large corpus (1.71×10^6) of simulated

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[1]The published Phyloformer weights are amino-acid–specific; we retrain its DNA variant on the same simulated corpus (Section 3.4).

sequence pairs; an exact-label OOD CODON simulation; two biological corpora (250 OrthoMaM mammalian (Scornavacca et al. 2019) and 168 1KITE arthropod (Misof et al. 2014) sub-alignments); and seven simulated SRH-violation (Naser-Khdour et al. 2019) stress tests.

On the all-taxa in-distribution GTR simulation, our best variant trained on squared error achieves $2.6\times$ lower MSE than IQ-TREE (Nguyen et al. 2015); on the matched panel, Phyloformer also wins MSE and correlation. A different variant, one trained for relative rather than absolute error, produces the best pair-encoder NJ trees in the all-taxa comparison, although JC remains best on correlation and topology. On biological data, ground-truth distance is unobservable: relative to per-MSA IQ-TREE GTR+Γ_4 estimates—a reference, not observable truth—JC outperforms every tested neural method in Pearson correlation (r), Spearman correlation (ρ), and normalized Robinson–Foulds distance (nRF), and has the highest MSA-level head-to-head win rate. Three adaptation attempts do not close the tested OrthoMaM ranking gap; for the GTR-trained pair encoders, reference-free CODON and the combined SRH condition reproduce the reversal (Section 4).

We organize this evidence around two contributors. Classically, without a rate calibration, pairwise statistics cannot separate the rate-scale gauge from elapsed time (Proposition 3); after unit-rate normalization, within a fixed irreducible, time-reversible process, JC preserves true-distance ranks and never overestimates true distance (Theorems 4 to 6). More generally, an exact squared-loss decomposition splits target risk into a representation-dependent information ceiling and a simulation-to-target conditional shift (Theorem 1): the oracle and FIREFLY probe the first, although FIREFLY also changes architecture and pretraining, and exact-label CODON and matched non-reversibility training probe the second. These probes identify neither population term, and we prove neither the required finite-L separation nor its persistence; no result makes JC target-optimal for arbitrary biological processes (Corollary 2; Appendices A, B.1).

2 Background

2.1 Distance estimation under CTMC

DNA evolution is modeled as a continuous-time Markov chain (CTMC) on $\Sigma = \{A, C, G, T\}$ with rate matrix $\mathbf{Q}$, where $Q_{ab} \geq 0$ for $a \neq b$, $Q_{aa} = -\sum_{b \neq a} Q_{ab}$, and $\mathbf{P}(t) = e^{\mathbf{Q}t}$. For irreducible $\mathbf{Q}$ (all states communicate), $\boldsymbol{\pi}$ is the unique positive stationary distribution, $\boldsymbol{\pi}^\top \mathbf{Q} = 0$; time reversibility requires detailed balance, $\pi_a Q_{ab} = \pi_b Q_{ba}$. Evolutionary distance is $d = \mu(\mathbf{Q})t$, where $\mu(\mathbf{Q}) = -\sum_a \pi_a Q_{aa}$; we use the conventional unit-rate normalization $\mu(\mathbf{Q}) = 1$, hence $d = t$. The full definitions and labeled normalization equations appear in Appendix A.1.

Under Jukes–Cantor (JC; Jukes and Cantor 1969), all substitution rates and nucleotide frequencies are equal. If p is the observed mismatch fraction, its ML distance is

$$\hat{d}_{\mathrm{JC}}(p) = -\tfrac{3}{4} \ln\bigl(1 - \tfrac{4}{3}p\bigr). \tag{1}$$

This zero-parameter correction accounts for unobserved multiple substitutions at the same site.

GTR uses six exchangeabilities and four equilibrium frequencies and is commonly paired with discrete-Gamma among-site rate variation (Felsenstein 1981); IQ-TREE fits GTR+Γ jointly to an MSA and extracts pairwise ML distances. JC and GTR impose stationarity, reversibility, and homogeneity (SRH: stable composition, detailed balance, and a common lineage process); violations can bias model-based distances and motivate data-driven alternatives (Naser-Khdour et al. 2019).

3 Methods

3.1 Architecture

We use a 664,129-parameter column-wise pair encoder that jointly embeds each aligned nucleotide pair, contextualizes alignment columns with a three-layer, four-head Transformer, mean-pools valid positions, and maps the result through a two-layer MLP to a non-negative distance. The architecture schematic, full equations, parameter accounting, and symmetry checks appear in Appendix N.1; Appendix T specifies the auxiliary-feature extension.

3.2 Training corpus

Primary simulated corpus. The corpus contains 448 Yule trees simulated under GTR+Γ and split at the tree level into $300/73/75$ training/validation/test trees; after filtering, the training partition supplies 5,697,363 leaf-pair examples, and four test trees with malformed alignments are excluded from the evaluation. Labels are generating-tree path lengths, not estimates from JC or IQ-TREE. Training distances have mean 0.31 and median 0.18, and only 3.6% exceed 1.4, leaving the high-divergence 1KITE regime sparsely represented. Appendix M gives the generator, filtering, split, and distance distribution.

SRH stress-test corpora. We additionally construct seven corpora spanning a clean baseline; separate violations of stationarity, reversibility, homogeneity, and rate constancy (heterotachy); CpG context dependence; and a combined condition. Each contains 10^4 MSAs of 14 taxa with strong rate heterogeneity ($\alpha = 0.5$); Appendix M.5 and Table 13 give the full conditions and generation details.

3.3 Model variants and training

We train seven variants spanning three losses and two sampling schemes under a common optimizer protocol; full variant and training details are given in Appendix Tables 18 and 19.

3.4 Baselines

Jukes–Cantor. We compute the mismatch fraction p over valid sites and apply Equation (1), defined for $p \in [0, \frac{3}{4})$. Saturated pairs are omitted from the common pair set for every method; Appendix U gives the details and the tree-reconstruction exception.

IQ-TREE GTR+Γ_4. We fit IQ-TREE 2 (Nguyen et al. 2015) independently to each MSA under a four-rate-class discrete-Gamma GTR model; its maximum-likelihood pairwise distances serve as both the IQ-TREE baseline and the

biological reference. Appendix M.7 documents commands and distance extraction.

Phyloformer-DNA. Because the published Phyloformer weights are amino-acid–specific, we retrain its DNA variant from scratch on the same simulated corpus using the published recipe, without a separate exhaustive hyperparameter sweep. This controls the training data; because architecture, loss, and optimization differ, we do not interpret the comparison as an architecture-only ablation. The retrained model attains the best matched-GTR MSE (0.019) and Pearson correlation ($r = 0.852$), confirming competent training (Appendix Table 10). Its $O(N^2L)$ memory restricts the all-taxa simulated comparison, but every biological sub-alignment has at most 30 taxa, so Phyloformer receives full taxon width on biological data. Appendix O gives the full reproduction and computational details.

FIREFLY. FIREFLY is a full-MSA Transformer with a phylogenetic-bias module in column attention. We evaluate its released broadly pretrained weights rather than retraining on our GTR corpus, so it changes both available MSA context and training distribution; we therefore treat it as a robustness probe, not a controlled architecture ablation, and attribute its behavior to neither factor alone. It runs at full taxon count on CODON but exhausts memory on the longest GTR alignments, explaining its absence from the all-taxa GTR panel of Table 1; Appendix O gives model and scaling details.

Log-det / paralinear. Log-det (Steel 1994; Lockhart et al. 1994) and paralinear (Lake 1994) distances are appendix-only auxiliary controls rather than primary baselines (Remark 10; Appendix S).

3.5 Evaluation protocol

Datasets. We evaluate on (i) the held-out GTR evaluation set (71 trees; 1,708,890 all-taxa pairs, or 126,120 pairs under the matched $\leq$ 30-taxon protocol—full width for $N \leq 30$ and a 30-taxon subsample otherwise); (ii) curated OrthoMaM v12 mammalian ortholog sub-alignments (Scornavacca et al. 2019) (250 sub-alignments, 50 genes, 8–30 taxa, mean $d \approx 0.18$); (iii) 1KITE arthropod ortholog sub-alignments (Misof et al. 2014) (168 sub-alignments, 106 genes, 8–30 taxa, mean $d \approx 1.42$, between the training distribution's 95th and 99th percentiles); (iv) seven SRH stress tests (10^4 MSAs each); and (v) an out-of-distribution codon corpus (71 trees, 355 nucleotide MSAs; Appendix M.3). Thus OrthoMaM and 1KITE probe low- and high-divergence biological regimes, respectively.

Reference status. GTR, SRH, and CODON pairs use exact generating-tree path lengths. Biological ground truth is unobservable, so OrthoMaM and 1KITE use per-MSA IQ-TREE GTR+Γ_4 maximum-likelihood distances: biological "agreement" therefore means agreement with that reference, not known accuracy on truth. Any reference misspecification is inherited by the comparison; Appendix W reports model-adequacy and codon-aware checks.

Metrics and aggregation. We report Pearson r, Spearman ρ, MSE, MAE, the scale ratio $\bar{\hat{d}}/\bar{d}$,[2] and nRF between NJ trees (Saitou and Nei 1987; Robinson and Foulds 1981); the oracle additionally reports R^2. Correlation, error, and nRF metrics are computed within each MSA and then averaged, whereas Scale is pooled across all pairs. Appendix M.8 gives full definitions, including nRF normalization and the appendix-only H2H statistic.

Fairness and inference. Except for explicitly captioned encoder-specific exclusions, every method uses the same valid pair set within each comparison. On biological data, JC and IQ-TREE use full alignments while neural inputs are capped at $L = 2000$; under uniform $L = 2000$, JC retains its Pearson/Spearman advantage over two tested pairwise variants (Appendix V). Bootstrap 95% intervals use 10^3 gene-level, not pair-level, resamples; significance uses paired Wilcoxon signed-rank tests with Holm–Bonferroni correction. Appendix M.8 gives the complete conventions.

4 Results

4.1 Neural methods win in-distribution MSE

Table 1 summarizes the all-taxa evaluation on 1.71×10^6 simulated test pairs, with all displayed methods evaluated on the same pair set. The best variant ($\mathcal{M}_{\mathrm{MSE+}}$, squared-error loss with bin-balanced sampling) achieves MSE = 0.034: $2.6\times$ lower than IQ-TREE (0.089) and $6.1\times$ lower than JC (0.208). For tree reconstruction, $\mathcal{M}^{\mathrm{MRE}}_{\log+}$ (bin-balanced log-distance loss, relative-error fine-tuned) produces the best pair-encoder NJ trees (nRF = 0.433), beating IQ-TREE (0.531) but not JC (0.420). Despite matching the nominal GTR+Γ generator, IQ-TREE has the largest nRF in this evaluation.

No single pair-encoder variant dominates all metrics: $\mathcal{M}_{\mathrm{MSE+}}$ minimizes MSE, $\mathcal{M}_{\log}$ (log-distance loss, uniform sampling) maximizes correlation, and $\mathcal{M}^{\mathrm{MRE}}_{\log+}$ yields the best pair-encoder NJ trees. Complete per-variant and distance-stratified GTR results appear in Appendix Table 9.

On the matched GTR evaluation under the $\leq$ 30-taxon protocol, Phyloformer achieves the lowest MSE (0.019) and highest correlation ($r = 0.852$); both full-MSA models improve on every pairwise variant in nRF, but neither passes JC (JC 0.381, FIREFLY 0.385, Phyloformer 0.391, best pairwise 0.393; Appendix Table 10). JC, Phyloformer, and the pair encoders use 126,120 pairs; FIREFLY uses 126,005 because its encoder drops a few unencodable sequences.

4.2 Out-of-distribution simulation: the neural advantage is model-specific

We design this regime to separate biological and reference-estimation noise from out-of-distribution effects. We generate sequences under a process the models were never trained on, reusing the GTR test-tree topologies and nominal alignment widths, with exact ground-truth distances.

Because a codon model acts on nucleotide triplets rather than as the four-state, site-independent process GTR assumes, it changes the generative process rather than merely

[2]The scale ratio $\bar{\hat{d}}/\bar{d} = (\sum_i \hat{d}_i)/(\sum_i d_i)$ is the mean predicted distance divided by the mean true (or, on biological data, reference) distance, pooled across all pairs; it measures systematic multiplicative bias in the distance scale—1 is unbiased, below 1 underestimation, above 1 overestimation.

Method	MSE	Pearson r	nRF
(a) In-distribution GTR, all taxa			
JC	0.208	**0.846**	**0.420**
IQ-TREE GTR+Γ	0.089	0.689	0.531
$\mathcal{M}_{\mathrm{MSE}+}$	**0.034**	0.806	0.465
$\mathcal{M}_{\log}$	0.038	0.823	0.445
$\mathcal{M}_{\log+}^{\mathrm{MRE}}$	0.039	0.816	0.433
(b) Out-of-distribution CODON, all taxa			
JC	32.37	**0.934**	**0.325**
FIREFLY	**30.20**	0.932	0.339
$\mathcal{M}_{\mathrm{MSE}}$	33.84	0.826	0.387
$\mathcal{M}_{\log+}^{\mathrm{MRE}}$	34.19	0.831	0.374

Table 1: Simulation reversal. (a) In-distribution GTR (71 all-taxa trees): pair encoders win MSE; JC wins r and nRF. (b) Out-of-distribution CODON (71 trees, 355 MSAs; exact patristic labels): every GTR-trained pair encoder trails JC; FIREFLY has lower MSE and near-matched r. Memory limits keep full-MSA baselines out of (a) (matched appendix). Each panel uses a common pair set and shows only the pairwise variants best on some metric, so displayed variants differ ($\mathcal{M}_{\mathrm{MSE}}$: squared-error, uniform; names in Table 18). CODON branches are $\approx 3\times$ the nucleotide scale (distances reach ≈ 6.4), so cross-panel MSE is not comparable; (b) drops 112 JC-saturated pairs. Per-variant, matched, and distance-stratified results in Appendix M.4. Bold marks the best per complete panel.

reparameterizing the training distribution, giving a controlled out-of-distribution check. We reuse the same 71 test-tree topologies with branch lengths rescaled $3\times$ (codon branch lengths are substitutions per codon) and simulate under a codon model, yielding 355 nucleotide-DNA MSAs with exact tree-path labels and no biological or reference-estimation noise. Full generator, replication, rounding, and distance-scale details appear in Appendix M.3.

Under this controlled shift, the ranking changes sharply. JC attains the best correlation ($r = 0.934$) and topology (nRF $= 0.325$), its MSE trailing only FIREFLY; all six headline GTR-trained pair-encoder variants are worse on every metric, despite beating JC by $\approx 6\times$ on in-distribution GTR MSE (Table 1). The matched comparison is sharpest: Phyloformer wins matched-GTR MSE (0.019) and correlation ($r = 0.852$), yet loses to the zero-parameter JC on CODON. FIREFLY—broadly pretrained and never fit to our GTR corpus—is the exception, attaining the lower MSE in both CODON protocols while nearly matching JC on correlation and trailing it only marginally on topology. For the GTR-trained pair encoders, this reversal is consistent with sensitivity to training-distribution mismatch rather than biological or reference-estimation noise alone; FIREFLY's differing representation and pretraining prevent attributing its robustness to either factor. Complete per-variant and matched results appear in Appendix Tables 11 and 12.

Method	Pearson r	Spearman ρ	Scale	nRF
(a) OrthoMaM				
IQ-TREE				
NJ	1.000	1.000	1.00	**0.190**
JC	**0.991**	**0.996**	0.77	0.198
$\mathcal{M}_{\log+}^{\mathrm{MRE}}$	0.910	0.917	1.33	0.287
Phyloformer	0.870	0.857	7.85	0.352
FIREFLY	0.963	0.952	1.14	0.256
(b) 1KITE				
IQ-TREE				
NJ	1.000	1.000	1.00	**0.395**
JC	**0.971**	**0.981**	0.38	**0.426**
$\mathcal{M}_{\log+}^{\mathrm{MRE}}$	0.754	0.729	0.55	0.604
Phyloformer	0.795	0.785	1.55	0.500
FIREFLY	0.893	0.886	1.09	0.485

Table 2: Biological summary. OrthoMaM (250 sub-alignments, 50 genes) and 1KITE (168 sub-alignments, 106 genes), with per-MSA IQ-TREE GTR+Γ_4 references. Pearson r, Spearman ρ, and nRF are averaged per MSA; Scale is pooled. JC and IQ-TREE use full alignments, whereas neural methods use $L = 2000$. Uniform $L = 2000$ preserves JC's Pearson/Spearman advantage over two tested pairwise variants; full-MSA baselines and topology are not re-evaluated (Appendix V). Full per-variant and H2H results appear in Appendix M.9.

4.3 Real data: JC matches the IQ-TREE reference more closely than neural methods

Table 2 summarizes the empirical reversal: the in-distribution neural advantage does not transfer, and JC agrees more closely with the reference than any tested neural method on every headline correlation and topology metric. Figure 1 (Section I) visualizes the cross-regime pattern for six pair encoders and Phyloformer-DNA.

On OrthoMaM (mammals, low divergence), JC's correlations far exceed those of the best pairwise neural variant ($\mathcal{M}_{\log+}^{\mathrm{MRE}}$), and its NJ trees essentially match the IQ-TREE reference while clearly beating every pairwise neural method on topology. On 1KITE (arthropods, high divergence), the gap widens—JC's correlation lead over the best pairwise variant grows sharply. Full-MSA methods narrow it, and FIREFLY is the strongest neural method on both corpora, but it still trails JC on correlation and tree topology (Table 2).

Scale behavior is consistent with, but does not isolate, training-distribution mismatch: pair encoders regress toward the training-distance regime, FIREFLY is near unity but changes representation and pretraining together, and Phyloformer is severely miscalibrated on OrthoMaM. Detailed ratios and reproduction checks appear in Appendix Tables 16 and 17 and Appendix E.

The reference choice raises a central question: *since JC is a restriction of GTR, is the apparent win partly an artefact of using a GTR-based IQ-TREE estimate as the proxy?* Parametric-bootstrap and codon-aware re-reference checks

Feature set	MSE	R^2
Single-site (3: JC, p, Ti/Tv)	1.002	0.48
All pairwise (8 features)	0.995	0.48
MSA only (5 features)	1.462	0.24
All 13 features	**0.454**	**0.76**
Full Transformer (664K params)	0.411	0.78

Table 3: Oracle MLP ablation on auxiliary features. Adding MSA context raises R^2 from 0.48 to 0.76 ($1.58\times$) and reduces MSE $2.2\times$.

narrow but do not flip the gap, while a non-GTR reference remains untested (Appendices W and F).

After post-hoc affine scale correction, JC retains lower pooled MSE than every tested neural method on both corpora; this is a scale/offset sensitivity analysis, not held-out calibration (Appendix U).

All 28 tested JC-versus-pair-encoder correlation comparisons are significant at $p < 10^{-5}$ (Holm–Bonferroni corrected).

4.4 Oracle ceiling analysis

To probe model capacity and information content, we train a small oracle MLP ($13 \to 256 \to 128 \to 64 \to 1$, ≈45K parameters) on the same 13 auxiliary features that our model receives, systematically ablating feature groups.

The results in Table 3 provide a finite-model empirical proxy for the information ceiling. All eight pairwise features—including k-mer divergences at multiple scales—achieve exactly $R^2 = 0.48$. The k-mers add nothing beyond what JC distance, p-distance, and the transition/transversion ratio already capture. Adding five MSA-level features (mean column entropy, fraction of invariant columns, mean and standard deviation of pairwise JC, number of taxa) raises R^2 $1.58\times$ to 0.76 and reduces MSE $2.2\times$. The MSA context disambiguates whether low sequence similarity reflects genuinely large distance (in a slowly-evolving tree) or substitution saturation (in a fast-evolving tree).

Our full Transformer ($R^2 = 0.78$) slightly exceeds the 13-feature oracle, suggesting that the tested pairwise architecture extracts nearly all information captured by these features. This finite-model result is consistent with a data bottleneck, but is not a direct estimate of the Bayes ceiling. It connects to the information bottleneck of Section 5: pairwise statistics are limited under model misspecification, and richer MSA-level context can expose additional information.

4.5 Robustness and boundary controls

On OrthoMaM, fine-tuning raises Pearson r from 0.910 to 0.945 but remains below JC's 0.991; held-out isotonic calibration and residual-over-JC regression likewise do not recover JC's rankings (Appendices J, K, and L). JC has the highest correlation in all seven fixed-training SRH tests (Appendix M.5). With training matched to the tested non-reversible process within $\gamma_{\text{eff}} \leq 0.23$, neural models reverse the comparison ($r \approx 0.90$ versus 0.81; Appendix M.6).

The tree-space-ruggedness scores of Pythia (Haag and Stamatakis 2025) also do not support a simple test-harder-than-train explanation (Appendix Z). At $L = 2000$, raw mismatch p remains ahead of the best pairwise neural variant in Spearman ρ on OrthoMaM (0.990 versus 0.948) and 1KITE (0.817 versus 0.733), showing that JC's logarithmic correction is unnecessary for the rank advantage and supporting a role for no learned training-distribution dependence (Appendix S). Across 50 OrthoMaM MSAs, disagreement among the seven variants tracks absolute IQ-TREE-reference error (mean $r = 0.64$) but is not calibrated predictive uncertainty (Appendix P).

5 Why does JC win? A theoretical analysis

5.1 Simulation-to-reference conditional shift

The empirical reversal has two distinct possible sources: the model input may omit information needed to determine distance, and the conditional mapping learned from simulations may differ from the mapping required at evaluation. We make this separation explicit and then connect each term to a targeted experiment.

Setup. For sequence length L, let $\mathcal{D}_L^0$ and $\mathcal{D}_L^\star$ be the simulation-training and target laws over a model input Z_L and distance label Y_L. Here Z_L is exactly the information supplied to the estimator: an aligned pair together with any auxiliary summary features for our variants, versus full-MSA context for a joint model.

The aligned columns enter our encoder through the joint nucleotide-pair embedding of Section 3.1. The label Y_L is the true path distance in simulation and the IQ-TREE GTR+Γ_4 reference on biological data.

Assume $Y_L \in [0, T]$, fix a bounded measurable version of m_L^0 on the target support (the sufficient support condition is $\mathcal{D}_{L,Z}^\star \ll \mathcal{D}_{L,Z}^0$), and define

$$
\begin{aligned}
m_L^0(z) &= \mathbb{E}_0[Y_L \mid Z_L = z], \\
m_L^\star(z) &= \mathbb{E}_\star[Y_L \mid Z_L = z], \\
\mathcal{R}_\star(f) &= \mathbb{E}_\star[(f(Z_L) - Y_L)^2].
\end{aligned}
$$

Theorem 1 (Target-risk decomposition under conditional shift)**.** *The target risk of the population-optimal simulation-trained squared-loss predictor m_L^0 decomposes as*

$$
\mathcal{R}_\star(m_L^0) = \underbrace{\mathbb{E}_\star[\mathrm{Var}_\star(Y_L \mid Z_L)]}_{\text{target Bayes risk for fixed } Z_L} + \underbrace{\mathbb{E}_\star[(m_L^0(Z_L) - m_L^\star(Z_L))^2]}_{\text{simulation–target conditional shift}}.
$$

Equivalently, $m_L^\star$ is the minimum-risk predictor using Z_L, and the excess risk of m_L^0 is exactly the conditional-shift term.

This is the standard squared-loss orthogonality identity; full conditions and extended consequences appear in Appendix B.1, while the equivalent identities and full proof appear in Appendix D.6.

Interpretation and scope. The two terms answer different questions. The conditional-variance term is the best attainable squared-loss risk once the representation is fixed: information absent from Z_L cannot be recovered by a larger predictor. The second term asks whether the simulation-optimal conditional map is also target-optimal on the inputs that occur at evaluation. It can remain positive even when the marginal distributions of Z_L overlap, because the same observed input can imply different expected labels under the two laws. Conversely, covariate shift in Z_L alone does not create this term when $m_L^0 = m_L^\star$ on the target support (Corollary 16).

The equality is a population, squared-loss statement: a finite trained network can add approximation, estimation, and optimization error. We therefore use the decomposition to organize mechanisms and experiments, not to claim that the displayed two terms numerically exhaust every finite-model error.

Term 1: information ceiling (oracle and FIREFLY). The first term is the irreducible squared-loss risk after fixing Z_L. Adding summaries that are deterministic functions of the same aligned pair may change finite-model parameterization but cannot add information in the population sense; MSA summaries or additional taxa can. If $Z_L^{\mathrm{ctx}} = (Z_L^{\mathrm{pair}}, C_L)$, the law of total variance gives

$$\mathbb{E}_\star[\mathrm{Var}_\star(Y_L \mid Z_L^{\mathrm{ctx}})] \le \mathbb{E}_\star[\mathrm{Var}_\star(Y_L \mid Z_L^{\mathrm{pair}})].$$

Our finite-model oracle is an empirical proxy, not a direct estimate of either Bayes risk: pairwise features reach $R^2 = 0.48$, whereas adding five MSA-level summaries raises R^2 to 0.76 (a $1.58\times$ increase) and lowers MSE $2.2\times$ (Table 3). On the biological corpora, FIREFLY closes approximately 35% and 67% of the best-pairwise-to-JC nRF gap on OrthoMaM and 1KITE, respectively—roughly half overall—without surpassing JC on topology (Table 2). FIREFLY changes both available MSA context and broad pretraining, so these data attribute its advantage to neither factor alone. The formal statement and conditions are Corollary 15 in Appendix B.3.

Corollary 2 (Positive-measure mismatch gives a target-error lower bound)**.** *Suppose there exist constants $\alpha > 0$ and $\gamma > 0$ such that*

$$\mathcal{D}^\star_{L,Z}\big(\{z : |m_L^0(z) - m_L^\star(z)| \ge \gamma\}\big) \ge \alpha.$$

Then

$$\mathcal{R}_\star(m_L^0) - \mathcal{R}_\star(m_L^\star) \ge \alpha\gamma^2.$$

Here α lower-bounds the target mass of the mismatched region and γ the conditional-map separation on it. The corollary is a fixed-L implication; a non-vanishing asymptotic floor additionally requires that α and γ persist with L, and the decomposition proves neither the finite-L separation nor its persistence. The CODON, 1KITE, and biological results are consistent with such finite-L separation but do not prove persistence as $L \to \infty$. A latent-regime mechanism and the full two-regime calculation appear in Appendix B.3 (Lemma 14).

Term 2: conditional shift (CODON). CODON removes the biological-reference ambiguity: it evaluates the GTR-trained models on out-of-distribution codon-model MSAs using true tree-path distances. Every pairwise neural variant trails JC simultaneously in MSE, correlation, and nRF (Table 1). The full-MSA FIREFLY exception is important: it attains lower MSE than JC but is slightly worse on correlation and topology. Thus CODON removes reference-estimation artefacts and gives reference-free evidence consistent with conditional shift for the pair encoders; without codon-matched retraining, it does not identify conditional shift as the sole cause. The complementary boundary case is controlled matched training: across the tested non-reversibility envelope, per-condition matched neural models recover and overtake JC in correlation (Appendix M.5, Table 15). This is consistent with Corollary 16: the shift penalty is contingent on mismatch, not an inherent penalty for neural estimation.

These probes are complementary rather than individually identifying: the oracle varies the input representation, FIREFLY changes representation, architecture, and pretraining together, CODON changes the test generator without matched retraining, and matched SRH controls one mismatch axis but is not a biological-data intervention. Together they triangulate the two terms without claiming to estimate either population expectation directly.

Rate-versus-floor ML consequence. If a correctly specified parametric ML distance estimator has target MSE at most C/L and the positive-measure separation persists with L-independent α, γ, Corollary 11 implies that its target MSE is eventually below $\mathcal{R}_\star(m_L^0)$. This one-sided rate-versus-hypothesized-floor result is independent of finite neural capacity and does not assert correct specification of the biological data, cover the log-MSE or MRE Bayes acts, or imply ML dominance under misspecification (full statement and proof in Appendices B.1 and D.8).

Supporting classical structure. Because $e^{(c\mathbf{Q})(t/c)} = e^{\mathbf{Q}t}$, the pairs $(\mathbf{Q}, t)$ and $(c\mathbf{Q}, t/c)$ are observationally identical before rate normalization (Proposition 3). After fixing a single irreducible, time-reversible, unit-rate $\mathbf{Q}$, the expected mismatch $p_{\mathbf{Q}}(t) = 1 - \sum_a \pi_a P_{aa}(t)$ is strictly increasing in t, so the increasing JC transform preserves true-distance ranks within that fixed process (Theorem 4); moreover $\hat{d}_{\mathrm{JC}}(p_{\mathbf{Q}}(t)) \le t$, with equality for all positive t only for JC (Theorem 5), and Theorem 6 gives a composition-dependent lower bound on the gap.

These results explain why a zero-training estimator can retain useful within-process rankings despite scale bias. They do not guarantee NJ, compare pairs generated under different $\mathbf{Q}$'s, or cover non-reversible processes. Full statements, quantitative bounds, and scope remarks are in Appendix A, with proofs in Appendix D; the analogous K-state Poisson corrector (e.g. $K = 20$ proteins) has the same underestimation structure (Appendix C, Theorem 17).

Synthesis. The decomposition characterizes the population simulation-trained predictor—richer context can lower its information ceiling and matched training its conditional-shift term—while the classical results characterize JC as a training-independent comparator with useful rankings within a fixed reversible process. Neither makes JC target-optimal on arbitrary biological processes.

6 Related work

Three threads frame our study.

Neural phylogenetics. DeepED introduced convolutional evolutionary-distance estimation from aligned DNA pairs (Liu et al. 2020), and Phyloformer extended neural distance estimation to full MSAs using axial self-attention (Nesterenko et al. 2025). Rosenzweig and Hahn (2025) showed that compact, 7,657-parameter S-networks can approximate classical distances on simulated data; matching a closed-form estimator in distribution therefore need not require a large model. Concurrent FIREFLY uses an MSA Transformer and reports gains over Phyloformer on three biological CRW intronic-RNA datasets (Gao, Lee, and Liu 2026). It does not compare JC, log-det, or another zero-training classical estimator on biological data; TN93+FastME is evaluated only on simulations. In our experiments, FIREFLY also mixes broad pretraining with full-MSA context, so we treat it as a robustness probe, not a controlled architecture ablation. Our oracle provides a complementary finite-model probe of selected MSA-level information (Section 4.4).

Empirical evaluation and model violation. Recent reviews identify limited biological evaluation as a central weakness of neural phylogenetics (Braichenko, Borges, and Kosiol 2025; Zhu, Wang, and Yang 2025). This matters because SRH violations are prevalent in empirical partitions (Naser-Khdour et al. 2019), saturation can erase pairwise signal (Manuel et al. 2025), and model adequacy is consequential for deep phylogeny (Kapli et al. 2023). Our controlled SRH boundary case is summarized in Section 4.5 and detailed in Appendix M.6.

Classical misspecification. Bias under misspecified pairwise models has long been characterized: Tajima and Nei (1984) analyzed JC/K2P bias under substitution asymmetry, Rzhetsky and Nei (1994) treated among-site rate variation, and Rzhetsky and Nei (1992) provided precedent for corrected-distance tree estimation outperforming misspecified likelihood. Log-det (Steel 1994; Lockhart et al. 1994) and paralinear (Lake 1994) distances instead target a strictly broader non-stationary, non-reversible class (see also Steel, Hendy, and Penny 1998). They are auxiliary controls here: we do not claim that JC dominates them. Broader neural-task context, baseline-reproduction details, and the precise scope of the classical results appear in Appendices H.1, O, and A.

7 Discussion

The experiments and theory organize the sim-to-real reversal around two separable contributors: information unavailable to the chosen representation and simulation-to-reference conditional shift (Theorem 1). Neural methods win in-distribution simulated MSE; JC agrees more closely than every tested neural method with the IQ-TREE GTR+Γ_4 biological reference—not observable truth—on correlation and topology. For the GTR-trained pair encoders, exact-label CODON removes biological and reference-estimation noise, while matched training reverses the comparison within the tested non-reversibility envelope. These finite-data findings are consistent with both terms but identify neither as the sole cause; persistent asymptotic mismatch remains a hypothesis.

Representation and training-distribution mismatch. Raw mismatch fraction p reproduces JC's rank advantage (Section 4.5), showing that JC's logarithmic correction is unnecessary for the advantage and supporting a role for no learned training-distribution dependence. Its same-$\mathbf{Q}$, population rank preservation is one robustness property, not a guarantee of NJ recovery or robustness across processes (Theorem 4). The oracle raises R^2 from 0.48 for pairwise features to 0.76 with MSA context, while the full Transformer reaches 0.78; the 0.02 gap is within run-to-run variation. This is finite-model evidence that the pairwise representation omits useful context, not a direct Bayes-ceiling estimate. FIREFLY is competitive with JC on CODON and is the strongest neural method on biological data, supporting richer-context evaluation. However, it changes both full-MSA representation and broad pretraining, so the source of its gains is confounded; it also trails JC on the evaluated biological correlation and topology metrics (Sections 4.2 to 4.4). Together, the evidence is consistent with contributions from both representation and training-distribution mismatch; neither is identified as the sole cause.

Extended theoretical scope, protein and foundation-model evidence, and follow-up evaluations appear in Appendices A, G, and H.

Limitations. Seven boundaries remain. (i) The SRH tests reach only $\gamma_{\text{eff}} \approx 0.23$; stronger non-reversibility ($\gamma_{\text{eff}} \gtrsim 0.3$), which requires a non-JC base $\mathbf{Q}$, is untested. (ii) The training corpus has mean $\bar{d} \approx 0.31$, so the high-saturation regime ($\bar{d} \gtrsim 1.4$, exhibited by 1KITE) is extrapolation for every tested neural variant. We did not conduct a per-method hyperparameter search; although the real-data reversal is shared across all eight neural configurations, substantial retuning or an amino-acid–native full-MSA model could narrow it. (iii) The seven training variants are pairwise, and Phyloformer plus FIREFLY provide nonexhaustive full-MSA coverage. (iv) The IQ-TREE GTR+Γ_4 reference is a parametric ML proxy, not truth; bootstrap and codon-aware re-reference checks narrow but do not eliminate this concern, and a non-GTR reference remains untested (Sections F and W). (v) Tree-quality comparisons use only NJ; alternative tree-inference procedures may shift them. (vi) Training simulations omit indels although biological alignments include them, potentially affecting the OrthoMaM gap. (vii) The 20-state protein setting is empirically untested; only the Poisson-theory generalization is provided (Theorem 17).

Practical recommendation. For DNA distance estimation on the datasets and references evaluated here, JC remains the pragmatic default among the tested methods; whether future richer-context, distribution-matched neural models overturn this is the natural next test.

Broader impact statement This work characterizes limitations of neural methods for a specific biological inference task. Its practical implication is cautionary: researchers should validate neural phylogenetic tools empirically before deployment. We foresee no negative societal impacts.

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